

Supplementary material of the article "Top-down control and species composition influence non-linearly the short-term response of experimental food webs to a nutrient pulse perturbation"

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S1 Food chain model

This section provides more information about the chemostat model presented in Box 1. We derived our hypothesis about the shape of the relationships between biovolume responses to a nutrient pulse perturbation and the top-down control at the pulse (*TC*) from a simple differential-equation food chain model. In a chemostat, the dynamics of nitrogen N , algal biovolume A , and rotifer biovolume R are described as follows:

$$\begin{cases} \frac{dN}{dt} = (N_0 - N) \cdot \delta - \frac{rNA}{H + N} \\ \frac{dA}{dt} = \frac{rNA}{H + N} - \frac{gAR}{K + A} - \delta A \\ \frac{dR}{dt} = \epsilon \cdot \frac{gAR}{K + A} - \delta R \end{cases}$$

with nitrogen inflow $N_0 = 80 \mu\text{mol } N \cdot L^{-1}$, dilution rate $\delta = 0.2 d^{-1}$, algal maximal growth rate $r = 0.5 d^{-1}$, half-saturation constant of the nitrogen uptake by algae $H = 10 \mu\text{mol } N \cdot L^{-1}$, conversion efficiency $\epsilon = 0.25$, rotifer maximal grazing rate g varying between 1 and 4 d^{-1} , and half-saturation constant of the rotifer functional response $K = 40 \mu\text{mol } N \cdot L^{-1}$.

The model parametrisation reflects general properties of plankton food webs (e.g. Raatz et al., 2018; Rosenbaum et al., 2019; Wojcik et al., 2021), it was not fitted using data from an existing food web. Furthermore, we chose a parametrisation enabling the investigation of systems under stable equilibrium conditions (i.e. fixed point).

Analytically, these solutions are:

$$\begin{cases} \frac{dR}{dt} = 0 \Leftrightarrow A^* = \frac{\delta K}{\epsilon g - \delta} \\ \frac{dN}{dt} = 0 \Leftrightarrow N^{*2} - (N_0 - H - \frac{rA^*}{\delta}) \cdot N^* - N_0 H = 0 \\ \frac{dA}{dt} = 0 \Leftrightarrow R^* = \frac{K + A^*}{gA^*} \cdot \left(\frac{rN^*A^*}{H + N^*} - \delta A^* \right) \end{cases}$$

N^* , A^* and R^* values determined the state of the system before the perturbation occurred, and served as a baseline to assess the effects of the perturbation. To generate a gradient of top-down control values, we varied the value of g between 1 and 4, which also affected the state at equilibrium (cf. N^* , A^* and R^* in the equations above). Finally, we computed the top-down control as $-\log_2(A_{t_{\text{pulse}}}/A^*)$.

For each g value, we let the simulation run for 10 days, perturbed the system by adding 4 times N_0 to N as done in our experiments, and let the system run again. To perturb the system, we stopped the solver (`scipy.integrate.odeint`, python), modified the value of N and reinitialised the solver to get the post-pulse dynamics. Finally we computed the average change in the dynamics per day over the 12 days after the pulse using the metric of overall ecological vulnerability, in line with the experimental methods (cf. *OEV* and *avgOEV* in the Methods, main text).

When the algal species were at capacity (e.g. Fig. S1.1a in absence of rotifers), the algae exploited heavily the nutrient pulse, because free nutrients were previously scarce and the algae were nutrient-depleted. Most of the nutrients were thus fixed by the algae (cf. total nutrients are well above the free nutrients). However, the algal response was relatively limited because algae could only grow from the pre-pulse to

the post-pulse carrying capacity. Finally, algal concentration declined towards the end of the simulation due to dilution and not predation.

When TC was larger (Fig. S1.1b-c), algal and rotifer responses were maximal, i.e. the overall changes in the concentrations in comparison to the equilibrium values were large. These two scenarios corresponded to the hump, i.e. the values of TC where the responses to the pulse changed from an increasing to a decreasing trend when TC increased. Algae were moderately nutrient-depleted and could benefit from the pulse (free nutrients around $4 - 5 \mu mol$ at equilibrium), allowing them to escape from the control of predators, as additional production could not be taken immediately by the predators.

Finally, when TC was high (Fig. S1.1d-e), algae were not nutrient-limited (cf. free nutrients above $5.5 \mu mol$ at equilibrium), they could only profit a bit from the pulse perturbation (see small difference between the curves of free nutrients and total nutrients), and their additional production was quickly consumed by the predators. Consequently, responses of algae and rotifers were smaller than in the three previous cases.

The illustrative timeseries also showed that TC was not directly proportional to the rotifer concentration at equilibrium. For instance, when $TC = 0.87$, the rotifer concentration was around $2.5 \mu mol$ (Fig. S1.1b), however when $TC = 2.57$, the rotifer concentration was below $2 \mu mol$ (Fig. S1.1e).

Note that this model does not include any information about the stoichiometry or the ecophysiology of algae species. Therefore, it does not consider the delay in growth that algae can experience when they are nutrient-depleted (cf. additional experiments

in Appendix S2).

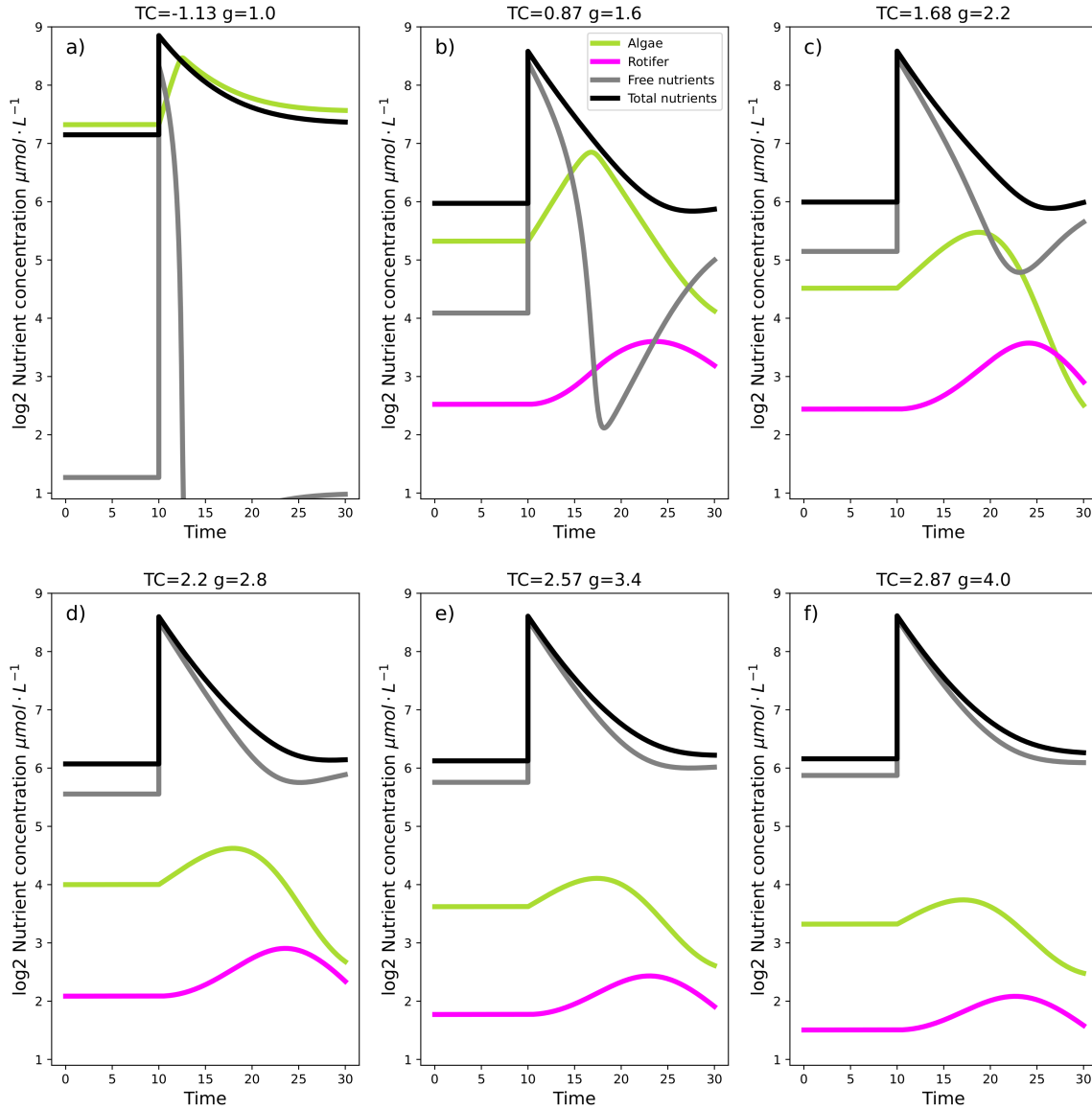


Figure S1.1: Six illustrative cases of dynamics of algae (green), rotifers (magenta), free nutrients (grey) and total nutrients (black) in $\log_2$ N-concentration unit ($\mu\text{mol} \cdot \text{L}^{-1}$). Total nutrients include the free nutrients available for algae, and the nutrients fixed in the algae and rotifers. Panels are ordered by increasing values of top-down control TC , which was partly determined by the grazing rate of rotifers g .

S2 Additional experiments

S2.1 Algal cultures in absence of rotifers

Biovolume response and carrying capacity

Before running the 24 algae-rotifer chemostats presented in the main text, we conducted chemostat experiments in absence of rotifers for each single algal species and the four algal species together (5 treatments x 2 replicates, i.e. 10 chemostats). Stock cultures, inoculation densities, the pulse perturbation, and experimental conditions of the chemostats were strictly the same as explained in the Methods in the main text.

These preliminary experiments served several purposes. First, we determined the carrying capacity of algal polycultures at approximately $4 \cdot 10^7 \mu m^3 \cdot mL^{-1}$, i.e. the mean total algal biovolume over the two replicates at the pulse, which enabled us to compute the top-down control in chemostats with rotifers (cf. Equation 2, main text). Second, these timeseries were used for estimating model parameters via Bayesian inference (see details in Appendix S4).

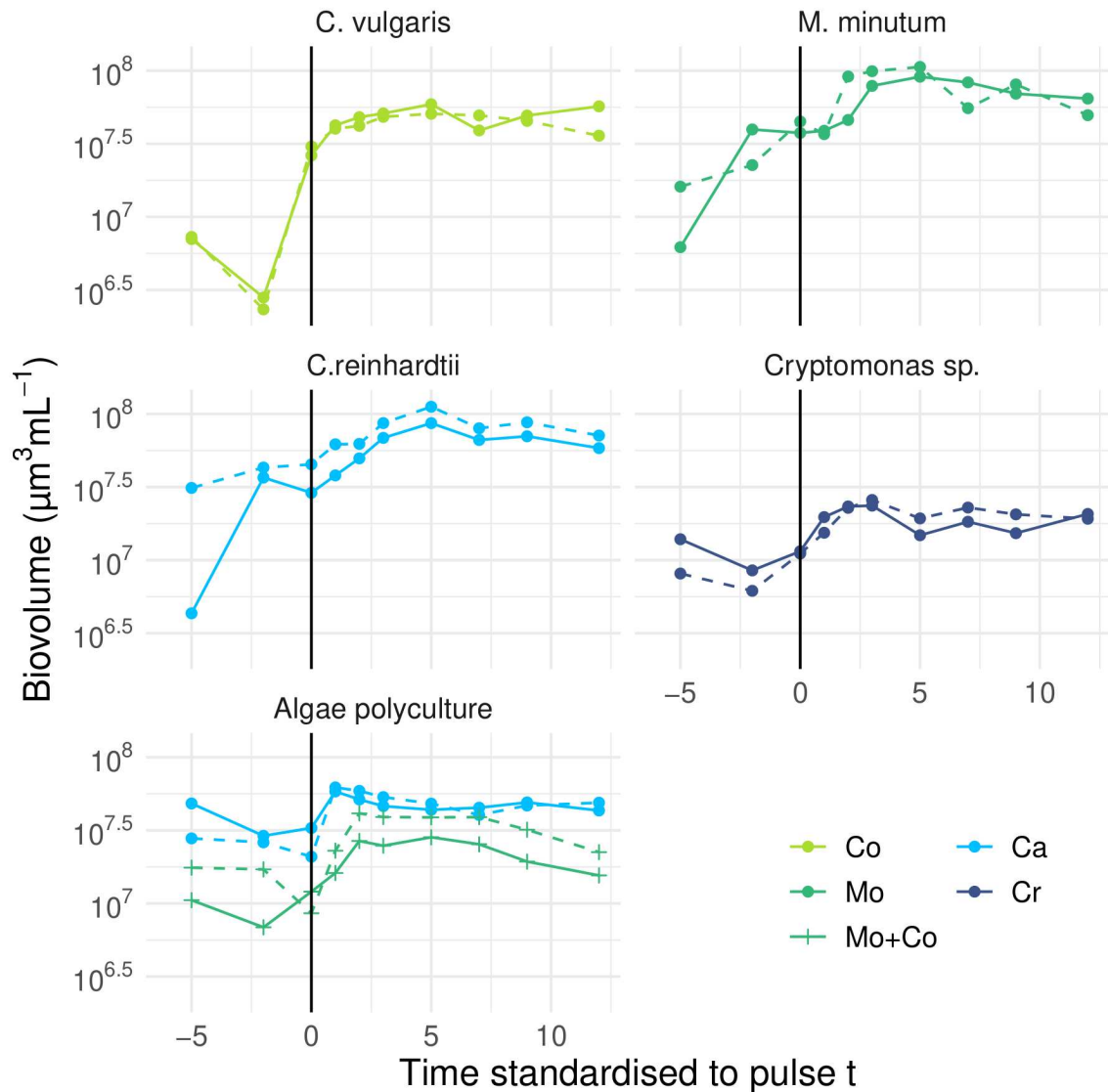


Figure S2.1: Biovolume timeseries in chemostats with monocultures of *C. vulgaris* (Co), *M. minutum* (Mo), *C. reinhardtii* (Ca), and *Cryptomonas sp.* (Cr) and with all the algal species together (polyculture). In all the panels, solid and dashed curves distinguish two replicates and the vertical black line marks the day of the pulse. In polycultures *M. minutum* and *C. vulgaris* could not be distinguished (cf. Appendix S3), thus their biovolumes were summed (Mo+Co), and *Cryptomonas sp.* biovolume was not displayed because it was several orders of magnitude lower than the biovolume of the other species ($< 10^5 \mu\text{m}^3 \cdot \text{mL}^{-1}$), but this species persisted until the end of the experiment.

Short-term numerical and stoichiometric response

To deepen our understanding of algal responses to the nitrogen pulse perturbation, we conducted short-term (nine days), semi-continuous experiments under identical temperature and light conditions as the chemostat experiments. The four algal species were reared in monocultures and in the absence of rotifers in 300-mL flasks containing 250 mL of WC medium. Six days after the inoculation, the same pulse perturbation as for the chemostat experiments was imposed (addition of $320 \mu\text{mol N} \cdot \text{L}^{-1}$, i.e. 4 times the amount of $N_0 = 80 \mu\text{mol N} \cdot \text{L}^{-1}$). Each algal monoculture was replicated twice, therefore we conducted a total of 8 experiments (4 species x 2 replicates).

The semi-continuous experiments were sampled on the second and the sixth day before the perturbation, and daily after the perturbation until the end of the experiment. On every sampling day 20% of the medium was renewed. For each experiment, two 5 mL-samples were filtered through 450°C pre-combusted glass fiber filters, and the carbon (C) and nitrogen (N) content per algal cell was measured thereafter using a gas chromatograph. In addition, cell densities and size spectra were measured using a CASY particle counter to calculate algal biovolume and cellular C and N-content.

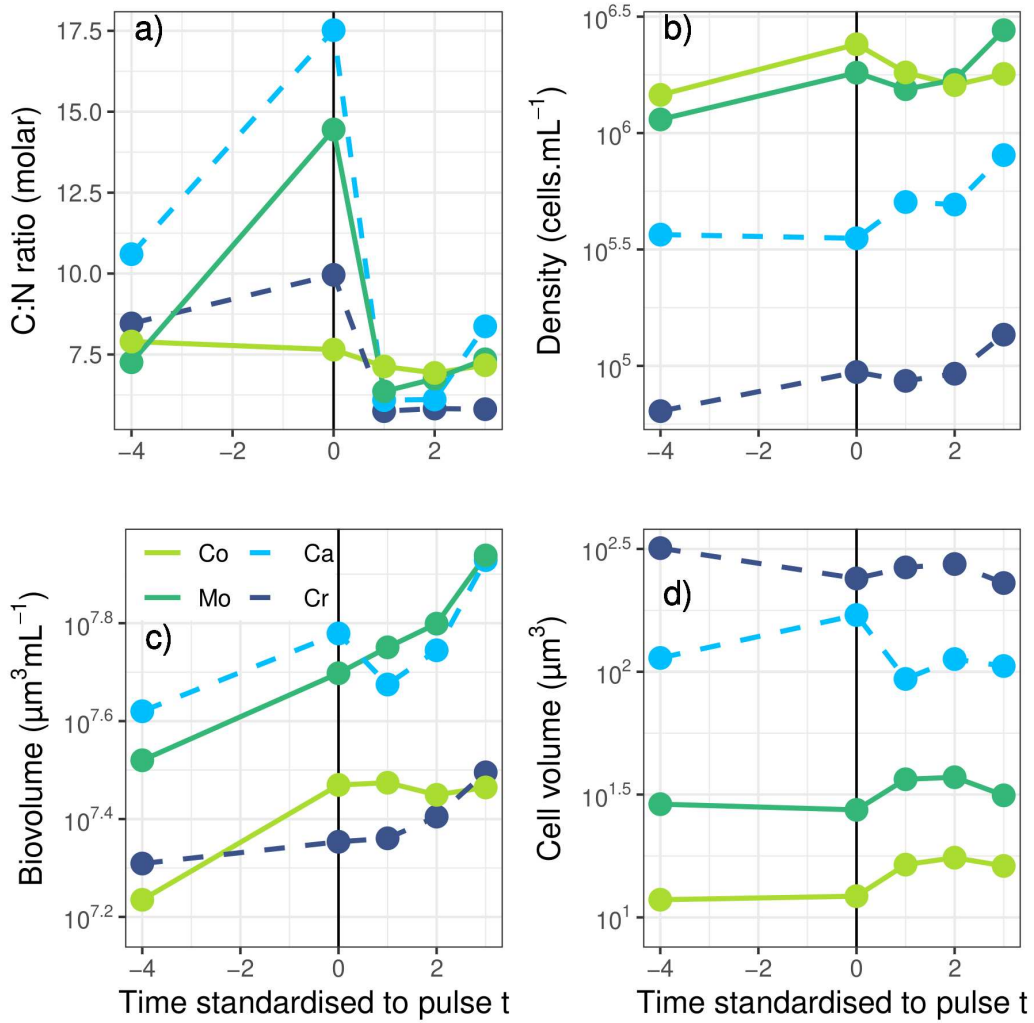


Figure S2.2: Timeseries of C:N ratios (a), cell densities (b), biovolumes (c) and cell volumes (d) for monocultures of *C. vulgaris* (Co in solid, light green curve), *M. minutum* (Mo in solid, dark green curve), *C. reinhardtii* (Ca in dashed, light blue curve) and *Cryptomonas* sp. (Cr in dashed, dark blue curve). Monocultures were cultivated semi-continuously and pulsed the day marked by a vertical, black line by increasing the nitrogen concentration from 80 to 400 $\mu\text{mol N} \cdot \text{L}^{-1}$.

These experiments showed species-specific ecophysiological differences. *C. reinhardtii* was the most nutrient-limited at the pulse (highest C:N ratio in Figure S2.2a), entered into a phase of cell division immediately after the pulse (Figure S2.2b), and its C:N ratio decreased faster than for the other species. The observed increase in biovolume after the pulse (Figure S2.2c) corresponded to a decrease in cell volume

which was compensated by a large increase in density (Figure S2.2b,d). *M. minutum* and *Cryptomonas sp.* responded similarly to *C. reinhardtii* but took up nutrients first (cf. decrease in C:N ratio and increase in cell volume in Figure S2.2a,d) and had a delayed increase in density (Figure S2.2b). *C. vulgaris* differed the most because it appeared as the least nutrient-limited prior to the pulse (C:N ratio remained around 7.5 in Figure S2.2a), and its biovolume remained constant after the pulse (Figure S2.2c) due to a decrease in density (Figure S2.2b) along with an increase in cell volume (Figure S2.2d).

S2.2 Feeding experiments using algae-rotifer pairs

We applied a presence-absence method adapted to rotifers (Seifert et al., 2014; Weithoff, 2005) to quantify the resource-dependent uptake of *C. vulgaris*, *M. minutum*, *C. reinhardtii* for the two dominating rotifer species *B. calyciflorus* and *Cephalodella sp.* (i.e. a total of 6 feeding experiments). To guarantee the rotifers' good physiological state prior to the feeding experiments, we fed rotifers at saturating food levels with algae in their exponential growth phase. For each feeding experiment, we ran five replicates without rotifers (controls) and five replicates with rotifers, for which rotifers were sieved beforehand through a mesh (40 μm for *B. calyciflorus* and 10 μm for *Cephalodella sp.*) to remove remaining food particles. We used six to seven density values of algal suspension spaced evenly along a $\log_2$ scale and we measured the exact starting algal density via light microscopy (Zeiss Axioskop 2, Germany). After 24 hours we estimated the final algal density (at least 600 cells counted) and

rotifer density (86 *B. calyciflorus* in 15 mL, i.e. $5.7 \text{ ind} \cdot \text{mL}^{-1}$ and 200 *Cephalodella* *sp.* in 3.5 mL, i.e. $57 \text{ ind} \cdot \text{mL}^{-1}$), and determined the feeding rates by computing the difference between the algal growth/decline without and with rotifers (Fig. [S2.3](#)).

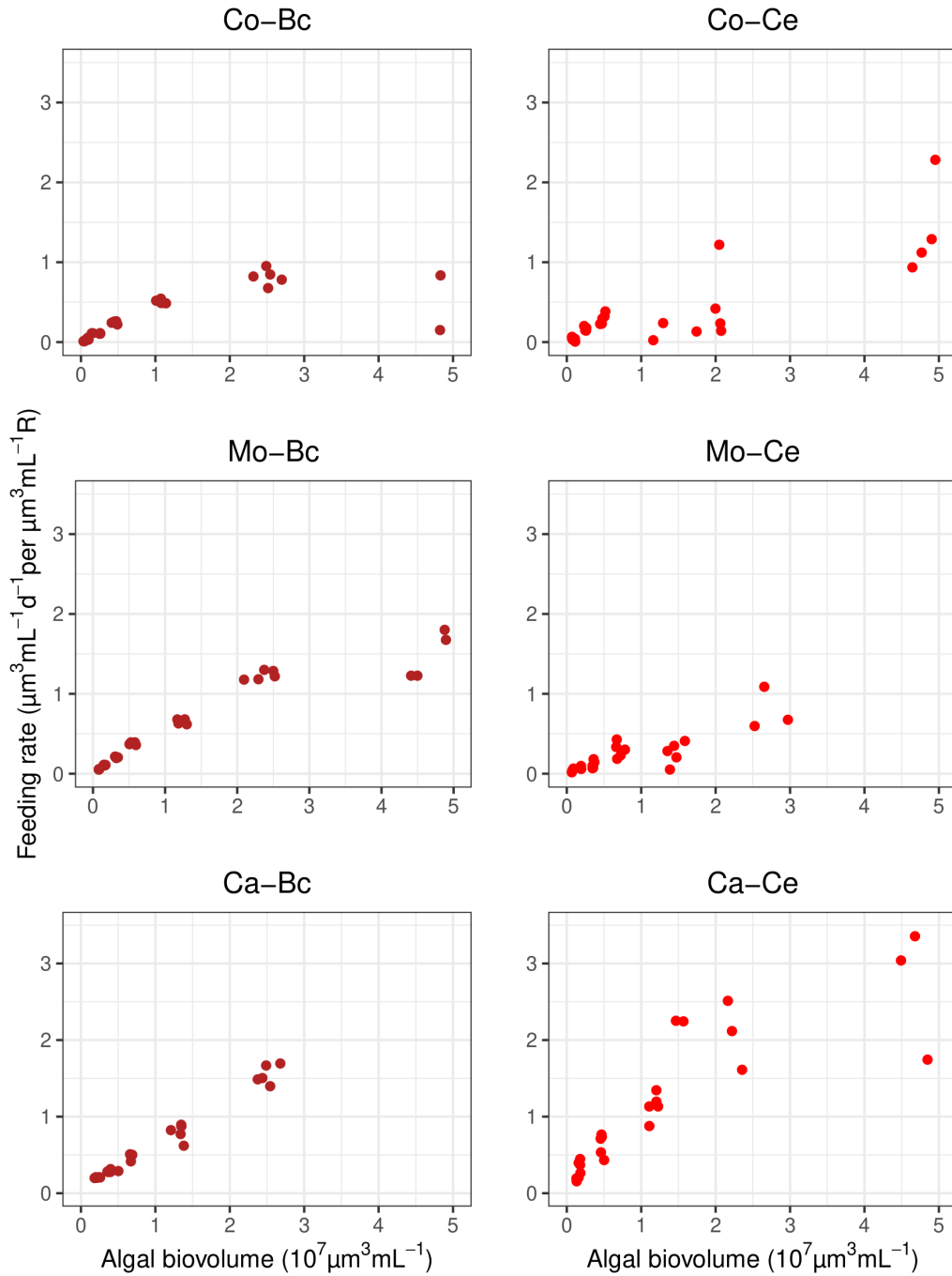


Figure S2.3: Rotifer feeding rates estimated for cultures of algae-rotifer pairs depending on the inoculated algal biovolume. All the combinations of the two rotifer species – *B. calyciflorus* (Bc in dark red) and *Cephalodella sp.* (Ce in red) – and three algal species – *C. vulgaris* (Co), *M. minutum* (Mo), and *C. reinhardtii* (Ca) – were investigated.

Generating a gradient of top-down control by varying inoculation densities

In addition to the 24 algal-rotifer chemostats where the TC gradient was generated based on trait changes, we conducted six chemostat experiments where we attempted to generate a TC gradient using the more traditional approach based on changes in the initial consumer-resource density ratio. In the six monocultures of *B. calyciflorus*, we varied the inoculation rotifer density along a gradient of values between 0.3 and $30 \text{ ind} \cdot \text{mL}^{-1}$. Stock cultures, the pulse perturbation, algal initial densities and experimental conditions of the chemostats were strictly the same as explained in the Methods in the main text.

We computed the algal and the rotifer responses to the nutrient pulse perturbation and the TC at the pulse (cf. methods in the main text) and observed the relationships between community responses and TC (Figure S2.4). We observed that the experiments could be lumped into two groups: experiments with low and high TC values at the moment of the pulse. Consequently, the density-based approach was unsuccessful to generate a gradient of TC values, which was a requirement to test our hypothesis about the shape of community biomass response– TC relationships.

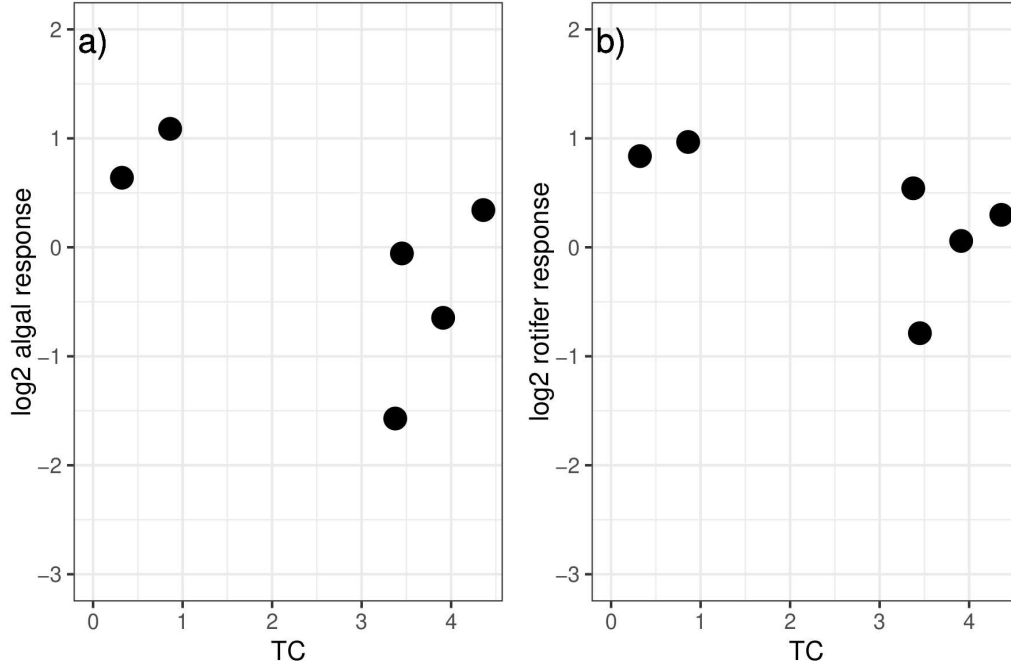


Figure S2.4: Relationships between a) $\log_2$ algal response ($avgOEVA$) or b) $\log_2$ rotifer response ($avgOEV_R$) and top-down control TC for six monocultures of *B. calyciflorus* which differed in rotifer inoculation density.

S3 Method to determine algal species composition

To determine the algal composition, we used algal monocultures (cf. S2) as references for flow cytometry (BD Accuri C6 Flow Cytometer, BD Biosciences) and three types of measurements: the forward scatter area (FCS-A, estimation of cell size), the filter for phycoerythrin (FL2-A, 585/40 nm) and the filter for chlorophyll (FL3-A, >670 nm).

Identifying *Cryptomonas* sp.

Cryptomonas sp. was larger than the other three species and the only one with phycoerythrin. We identified a particular ellipsoid area in the $\log_{10}$ - $\log_{10}$ plane FL2-

A – FSC-A in *Cryptomonas sp.* monoculture, which is defined as follows:

$$x^2 + \frac{271}{77} \cdot y^2 - 9 \cdot x - \frac{33361}{770} \cdot y < -152.5 \quad (\text{S3.1})$$

where x is $\log_{10}$ FL2-A and y is $\log_{10}$ FSC-A.

Identifying *C. reinhardtii* and small algae

We observed that *C. reinhardtii* had a higher chlorophyll signal per cell ($\log_{10}$ FL3-A, denoted as x in the following) and was larger (size $\log_{10}$ FSC-A, denoted as y in the following) than *C. vulgaris* and *M. minutum*. Yet, their respective x and y values could overlap, so we used a 2-dimensional density-based linear peak separation (Figure S3.1B) as follows:

1. We excluded the cells that were too small to be algal cells ($y < 5$).
2. In the chlorophyll histogram using class intervals (bins) of width 0.05 (Figure S3.1D), we determined the local maximum of *C. reinhardtii* x_{blue} for all values $x > b$ and the local maximum of *M. minutum* and *C. vulgaris* x_{red} for all values $x < b$ (cf. Figure S3.1D). The value $b = 5.76$ corresponded to the mean value between the temporal means of x for *C. reinhardtii* and *M. minutum*.
3. For all cells within the peak zone $x = x_{\text{red}}$, i.e. the cells in the class interval containing the value x_{red} and the class intervals before and after the class interval containing the value x_{red} , (see the three class intervals coloured in red in Figure S3.1D), we obtained the corresponding size histograms for which we

calculated the local maximal y_{red} (Figure S3.1A). We did the same for x_{blue} and y_{blue} (Figure S3.1C).

4. We established the equation of the line h passing through the two points with coordinates $(x_{\text{red}}, y_{\text{red}})$ and $(x_{\text{blue}}, y_{\text{blue}})$ in the x - y plane (Figure S3.1B).
5. We calculated the separation point y_b lying on h at $x = b$ (Figure S3.1B), and determined the expression of the line s passing through the point (b, y_b) and being orthogonal to h . Points on the left of s are cells categorised as *M. minutum* and *C. vulgaris*, whereas points on the right are categorised as *C. reinhardtii* cells.

This method did not allow to distinguish *M. minutum* from *C. vulgaris* unambiguously in all experiments because we could not establish a clear criteria regarding their size differences. In many experiments, their sizes changed shortly after the pulse and overlapped. Unfortunately, this observation was quite common and observed in previous work for certain algal species (e.g. Massie et al. (2013)), thus we considered the two species as one morphological group. In addition, this method may lack accuracy when x_{red} became larger than b . However, this happened rarely, i.e. shortly after a nutrient pulse in some chemostats, thus this method generally provides a good estimate of the algal composition in all the experiments.

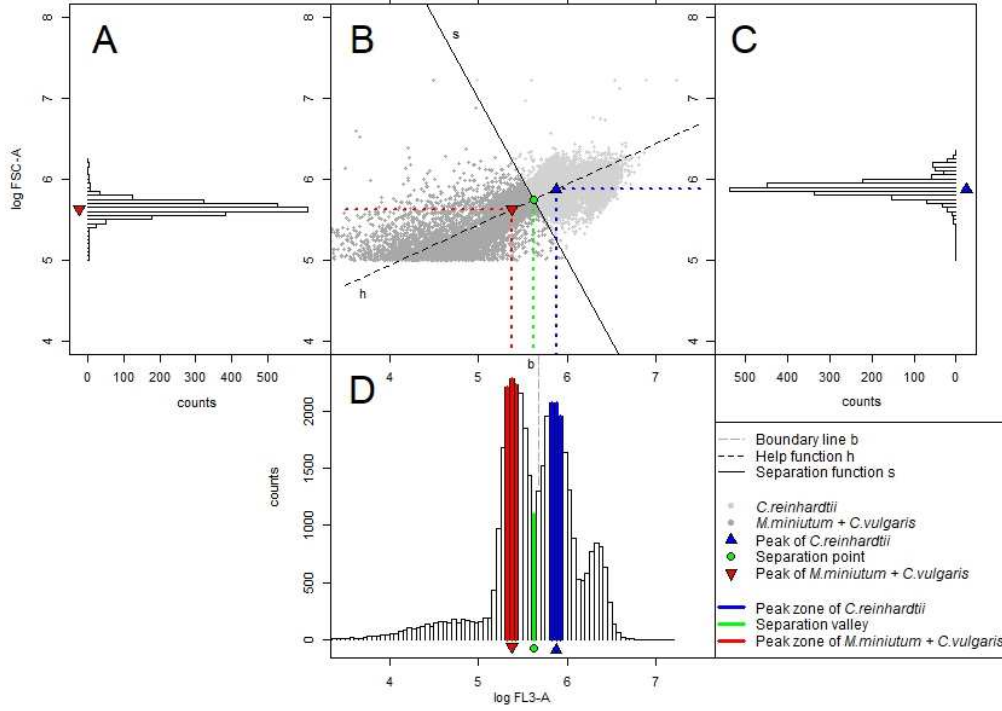


Figure S3.1: Illustration of the method to separate *C. reinhardtii* from *C. vulgaris* and *M. minutum* depending on two parameters obtained by flow cytometry analysis: $\log_{10}$ -transformed chlorophyll pigmentation (FL3-A, x-axis) and $\log_{10}$ -transformed cell size (FSC-A, y-axis). A and C: Histograms of counted cells along the y-axis representing size when $x = x_{\text{red}}$ for *C. vulgaris* and *M. minutum* (A), and $x = x_{\text{blue}}$ for *C. reinhardtii* (C). B: Scatter plots in x-y plane, with a helper function *h* passing through three points: $(x_{\text{red}}, y_{\text{red}})$, $(x_{\text{blue}}, y_{\text{blue}})$, and (b, y_b) , and the separation function *s*, which delimits the *C. vulgaris* and *M. minutum* cells on its left side from *C. reinhardtii* on its right side. D: Histogram of counted cells along the x-axis of chlorophyll pigmentation for all the species, where *b* is the the lowest boundary of chlorophyll pigmentation for *C. reinhardtii*, which separates the two peaks x_{red} and x_{blue} corresponding to the value of pigmentation with the most counted cells for *C. vulgaris* and *M. minutum* together, and *C. reinhardtii*, respectively.

S4 Bayesian inference

We first attempted to estimate algal and rotifer biovolume dynamics at the species level, however we failed to obtain a satisfactory fitting due to several reasons. First, two algal species could not always be distinguished experimentally without ambiguity (cf. Appendix S3), thus their respective dynamics were difficult to predict accurately. Second, we were mostly interested in the trophic level responses to the nutrient pulse, for which model errors of individual species were accumulated, thereby reducing the confidence in the biovolume predictions at the trophic level. Instead, when directly fitting the trophic level biovolume dynamics, the fitting was satisfactory and more in line with the food chain model onto which we based our theoretical hypothesis (cf. Box 1).

We used the following chemostat model to describe the dynamics of nitrogen $N(t)$ (in $\mu\text{mol } N \cdot L^{-1}$), and of trophic level biovolume of algae $A(t)$ and rotifer $R(t)$ (both in $\mu\text{m}^3 \cdot \text{mL}^{-1}$):

$$\begin{cases} \frac{dN}{dt} = (N_0 - N)\delta - \frac{1}{\epsilon_A} F_A(N) A \\ \frac{dA}{dt} = F_A(N) A - F_R(A) R - \delta A \\ \frac{dR}{dt} = \epsilon_R F_R(A) R - mR - \delta R \end{cases} \quad (\text{S4.1})$$

where nitrogen inflow N_0 is fixed at $80 \mu\text{mol} \cdot L^{-1}$, the dilution rate δ equals $0.2 d^{-1}$, ϵ_A (in $\mu\text{m}^3 \cdot \text{mL}^{-1} / \mu\text{mol } N \cdot L^{-1}$) is the conversion factor between nitrogen concentration and algal biovolume, ϵ_R (dimensionless) is the conversion factor between algae and rotifers, $F_A(N)$ is the algal nitrogen uptake, $F_R(A)$ is the rotifers' feeding rate, and m is the rotifers' natural mortality rate (in d^{-1}).

The nitrogen uptake by algae was described by Monod kinetics

$$F_A(N) = \frac{b N}{N + K} \quad (\text{S4.2})$$

with maximum growth rate b in d^{-1} and a half-saturation constant K in $\mu\text{mol } N \cdot L^{-1}$.

The rotifer feeding rate was modelled with a type II functional response

$$F_R(A) = \frac{a A}{1 + a h A} \quad (\text{S4.3})$$

with attack rate a in $mL/\mu m^3 \cdot d$ and handling time h in d .

We fitted model predictions to the observed, experimental timeseries data using Bayesian inference for estimating posterior probability distributions of all the parameters, i.e. K , b , ϵ_A , a , h , ϵ_R and m (Rosenbaum and Fronhofer, 2023). Ideally, all the treatments and chemostats would be fitted in a joint statistical model, which informs all the parameters simultaneously. Since this was computationally infeasible, we split up the fitting in several statistical submodels via Bayesian updating (cf. Table S4.1). Bayesian inference requires choosing prior distributions, thus when a parameter could not be informed by posterior distributions of a previous model, we chose informative priors from the literature.

First, we used data from two sets of preliminary experiments (see Appendix S2):

- (i) chemostats with only single algal species or all the four algal species and no rotifers (two replicates per single algal species and the four algae together, run under the same conditions as described in the main text), and (ii) feeding experiments of 6

combinations of algae-rotifer pairs (the rare species *Cryptomonas sp.* and *Lecane sp.* were excluded).

We fitted the first set of experiments in one joint model, but allowing algal species-specific parameters $\epsilon_{A,i}$, b_i and K_i ($i = 1, 2, 3, 4$), with priors adapted from Raatz et al. (2018) and Rosenbaum et al. (2019), and obtained posterior distributions for these parameters. For the second set of experiments we allowed algae-rotifer pairwise parameters a_{ij} and h_{ij} , with vague priors chosen through prior predictive checks (Wesner and Pomeranz, 2021), and obtained their posterior distributions. In this case the fitting occurred at the end of each feeding experiment, thus we accounted for resource depletion by computing predictions with an ODE model (Rosenbaum and Rall, 2018):

$$\frac{dA_i}{dt} = -F_{R,ij}(A_i) \cdot \overline{R_j} \quad (\text{S4.4})$$

Rotifer density was not modelled dynamically, but was kept constant at the mean rotifer density $\overline{R_j}$ of initial ($t = 0$) and measured final ($t = 1$) density. We estimated feeding rates from the difference in final abundances between the predator treatment and a predator-free control treatment $A_{i,E} = A_{i,C}(1) - A_i(1)$. A few observations with negative $A_{i,E}$ were omitted (8 in total). All the algal and rotifer densities were converted to $\mu\text{m}^3 \cdot \text{mL}^{-1}$ using factors $1\,409\,094 \mu\text{m}^3 \cdot \text{ind}^{-1}$ for *B. calyciflorus* and $73\,843 \mu\text{m}^3 \cdot \text{ind}^{-1}$ for *Cephalodella sp.*, and $288 \mu\text{m}^3 \cdot \text{cell}^{-1}$ for *C. reinhardtii*, $25.86 \mu\text{m}^3 \cdot \text{cell}^{-1}$ for *M. minutum* and $10.45 \mu\text{m}^3 \cdot \text{cell}^{-1}$ for *C. vulgaris*, such that the estimated parameters a_{ij} and h_{ij} have the correct units.

In a next step, we used the new posterior distributions as priors to estimate

parameters from (iii) 12 chemostat experiments of the main text including only one rotifer species. Here, we used trophic level-combined algal biovolume A and estimated trophic-level based parameters ϵ_A , b and K , but allowed rotifer-specific parameters a_j , h_j , $\epsilon_{R,j}$ and m_j , with priors for $\epsilon_{R,j}$ and m_j adapted from Raatz et al. (2018) and Rosenbaum et al. (2019).

In a last step, we used this new posterior information as priors for (iv) 12 chemostat experiments of the main text including all the three rotifer species (using trophic level-combined biovolume A and R), to obtain posterior distributions of the trophic level-based final estimates for all the parameters.

We followed a trajectory matching approach to fit model predictions to observed biovolume timeseries and used lognormal distributions to quantify residuals in the likelihood function (Rosenbaum and Fronhofer, 2023). We modelled all the chemostats' and species' initial biovolumes as free parameters informed by measurements, that were estimated simultaneously with the model parameters. In other words, all the chemostat timeseries replicates were fitted with an individual trajectory, which shares the same model parameters but only differs in initial conditions. Although nutrient concentration was not experimentally measured, we were able to model $N(t)$ dynamically as a latent variable by including initial nutrient concentrations as free parameters (Rosenbaum et al., 2019).

Since process errors can accumulate over time, it was generally difficult to achieve a good model fit for the whole biovolume timeseries with purely deterministic predicted trajectories. While theoretically this issue can be solved with state-space models (Auger-Méthé et al., 2021), we used here species' biovolumes at the time of the pulse

($t = 0$) as latent states, i.e. additional free parameters. This implies that trajectories before and after the pulse were fitted separately, with after-pulse initial biovolumes in $t = 0$ as free parameters being informed by the pre-pulse predicted trajectory in $t = 0$, and their discrepancies modelled as process errors.

(i) Algal cultures without rotifer

A_1	A_2	A_3	A_4	$\rightarrow \epsilon_{A,i}, b_i, K_i$
$A_1, A_2, A_3 + A_4$				

(ii) Feeding experiments

-	R_1, A_2	R_1, A_3	R_1, A_4	$\rightarrow a_{ij}, h_{ij}$
-	R_2, A_2	R_2, A_3	R_2, A_4	
-	-	-	-	

(iii) Rotifer monocultures

$R_1, A_1 + A_2 + A_3 + A_4$	$\rightarrow \epsilon_A, b, K, a_j, h_j, \epsilon_{R,j}, m_j$
$R_2, A_1 + A_2 + A_3 + A_4$	
$R_3, A_1 + A_2 + A_3 + A_4$	

(iv) Rotifer polycultures

$R_1 + R_2 + R_3, A_1 + A_2 + A_3 + A_4$	$\rightarrow \epsilon_A, b, K, a, h, \epsilon_R, m$
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Table S4.1: Fitted models (i)–(iv) and lists of parameters which were estimated from them. Indices i describe algae-specific and indices j describe rotifer-specific parameters, omitted indices describe trophic level-combined parameters. Every frame describes an experimental setup with the involved species. Species separated by a comma means that the statistical model included species-level biovolumes, while a plus symbol describes aggregated biovolumes.

	Median	sd	q05	q95
Chemostat	$\log(\epsilon_A)$ [in $\mu m^3 \cdot mL^{-1} / \mu mol N \cdot L^{-1}$]			
Monocultures	12.546	0.049	12.466	12.627
Polycultures	12.311	0.092	12.162	12.466
Chemostat	b [in d^{-1}]			
Monocultures	1.011	0.132	0.791	1.218
Polycultures	0.889	0.128	0.715	1.125
Chemostat	$\log(K)$ [in $\mu mol N \cdot L^{-1}$]			
Monocultures	0.791	0.469	0.186	1.708
Polycultures	0.697	0.456	0.167	1.597
Chemostat	ϵ_R [dimensionless]			
<i>B. calyciflorus</i>	1.328	0.177	1.078	1.654
<i>Cephalodella sp.</i>	1.421	0.348	0.93	2.058
<i>Lecane sp.</i>	0.993	0.337	0.548	1.634
Polycultures	2.483	0.35	1.943	3.095
Chemostat	$\log(h)$ [in d]			
<i>B. calyciflorus</i>	-1.119	0.723	-2.55	-0.21
<i>Cephalodella sp.</i>	1.993	0.264	1.549	2.415
<i>Lecane sp.</i>	-1.939	0.724	-3.235	-0.885
Polycultures	-0.929	0.932	-2.538	0.458
Chemostat	$\log(a)$ [in $mL/\mu m^3$]			
<i>B. calyciflorus</i>	-17.525	0.161	-17.791	-17.262
<i>Cephalodella sp.</i>	-15.919	0.747	-17.06	-14.607
<i>Lecane sp.</i>	-17.899	0.384	-18.493	-17.235
Polycultures	-18.825	0.157	-19.069	-18.558
Chemostat	m [in d^{-1}]			
<i>B. calyciflorus</i>	0.008	0.011	0.001	0.034
<i>Cephalodella sp.</i>	0.02	0.028	0.001	0.085
<i>Lecane sp.</i>	0.259	0.074	0.156	0.396
Polycultures	0.004	0.006	0	0.018

Table S4.2: Median, standard deviation, 5th-quantile and 95th-quantile estimates of parameters predicted via Bayesian inference. Algal-related parameters ϵ_A , b and K were estimated for the 12 rotifer monocultures together and the 12 polycultures, while rotifer-related parameters ϵ_R , h , a and m were estimated for the monocultures of *B. calyciflorus*, of *Cephalodella sp.* and of *Lecane sp.* separately, and for the polycultures.

S5 Supplementary results

Data used for testing hump-shaped relationships

Treatment	Chemostat	TC	$\log_2(avgOEV)$	
			A	R
Mono	Ce2	-0.06	-1.29	-1.03
	Le4	0.13	-1.29	0.87
	Le1	0.23	-0.74	1.06
	Ce4	0.29	-1.29	-0.67
	Ce3	0.54	-0.76	0.40
	Ce1	0.66	-0.62	-1.79
	Le2	1.05	0.07	-0.76
	Bc4	1.85	0.14	1.43
	Le3	2.26	0.50	0.94
	Bc2	2.64	-0.74	-0.89
	Bc1	2.74	-2.40	-2.00
	Bc3	3.34	-1.25	-2.40
Poly	002 (13d)	0.65	-0.56	-1.56
	007 (13d)	0.98	0.45	-1.40
	001 (13d)	1.00	-1.32	-1.00
	006 (9d)	1.27	-0.60	-0.83
	010 (9d)	1.59	-0.14	1.40
	011 (9d)	1.94	-0.38	0.82
	004 (9d)	2.44	0.56	0.36
	005 (9d)	2.49	-0.42	-0.97
	012 (9d)	2.83	0.24	0.62
	003 (13d)	3.01	-0.10	0.90
	008 (13d)	3.02	-1.09	-1.74
	009 (13d)	3.25	-1.79	-1.51

Table S5.1: Values of top-down control TC at the pulse, and responses ($\log_2(avgOEV)$) of algae (A) and rotifers (R) for each of the 24 chemostats. Chemostats are ordered by ascending TC values for the two treatments, i.e. rotifer monocultures (Mono) and polycultures (Poly), separately. For rotifer monocultures, chemostats are named after the rotifer present and a digit for each replicate (from 1 to 4), and for rotifer polycultures, chemostat names were associated to a replicate number and we precised on which day after the inoculation the pulse occurred (9th or 13th day). Biovolume timeseries of chemostats in bold font are presented in Figure 3 in the main text. All the values are rounded at the second decimal.

Summary statistics quadratic regressions

Table S5.2 provides all the summary statistics of the regressions performed in Figure 2.

$\log_2(\text{avgOEVA}) \sim a + bTC + cTC^2$ (Figure 2a)				
Treatment	Parameter	Coefficient	se	$P(c < 0)$
All		$n = 24, r^2 = 0.33$		
	a	-1.40	0.37	
	b	1.59	0.57	
	c	-0.47	0.17	1.00
Monocultures		$n = 12, r^2 = 0.42$		
	a	-1.38	0.44	
	b	1.74	0.80	
	c	-0.56	0.25	0.98
Polycultures		$n = 12, r^2 = 0.28$		
	a	-2.09	1.46	
	b	2.26	1.72	
	c	-0.60	0.43	0.94
$\log_2(\text{avgOEV}_R) \sim a + bTC + cTC^2$ (Figure 2b)				
Treatment	Parameter	Coefficient	se	$P(c < 0)$
All		$n = 24, r^2 = 0.19$		
	a	-0.82	0.65	
	b	1.44	0.97	
	c	-0.49	0.28	0.96
Monocultures		$n = 12, r^2 = 0.33$		
	a	-0.52	0.79	
	b	1.42	1.48	
	c	-0.59	0.46	0.95
Polycultures		$n = 12, r^2 = 0.43$		
	a	-4.77	2.14	
	b	5.58	2.52	
	c	-1.39	0.63	0.98

Table S5.2: Summary statistics of second-degree polynomial regressions of algal or rotifers response and top-down control at the day of the pulse TC . Regressions were performed for each treatment (rotifer monocultures and rotifer polycultures) separately or together (All).

Experimental biovolume timeseries

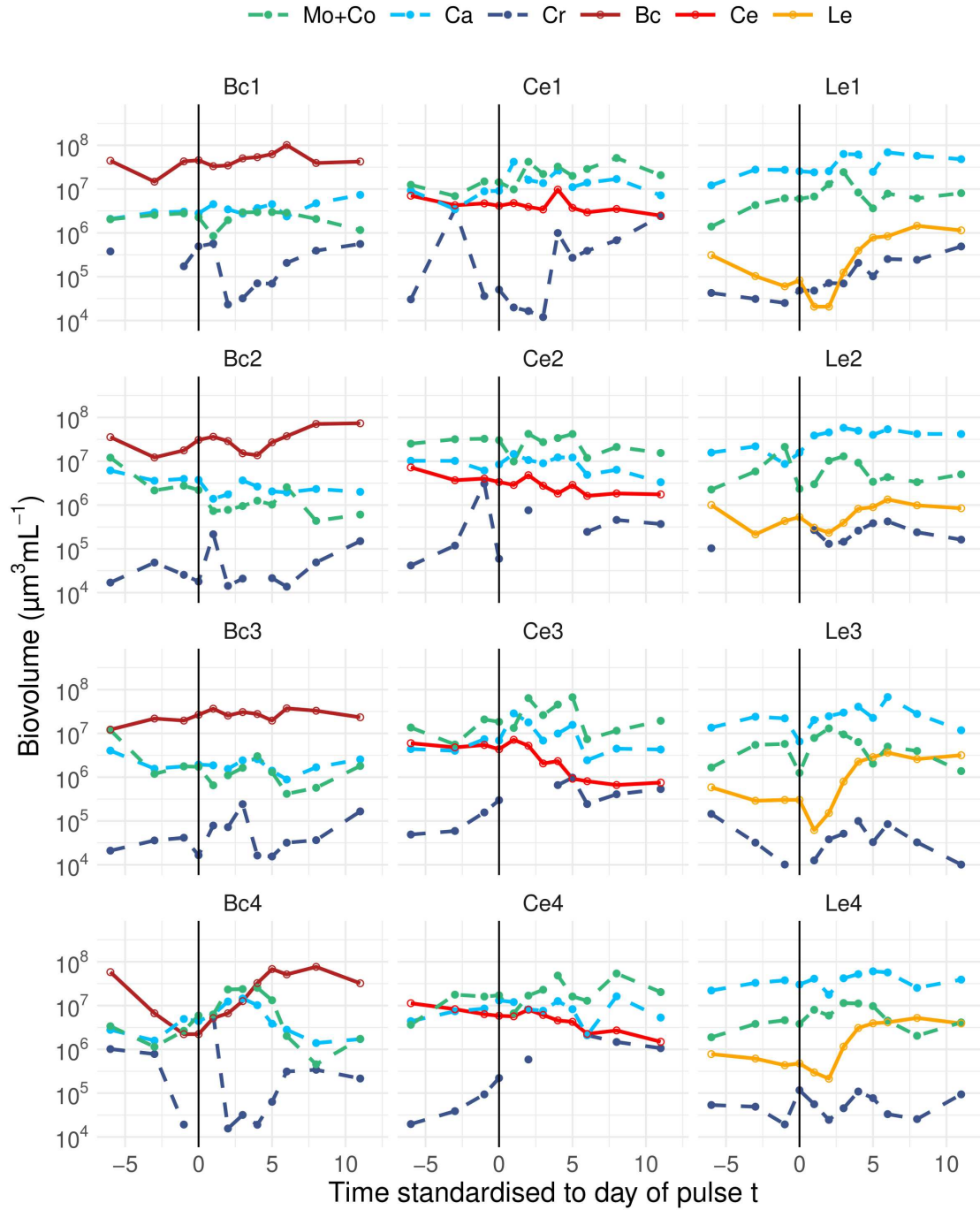


Figure S5.1: Biovolume timeseries of rotifer monocultures of *B. calyciflora* (Bc1-4), *Cephalodella sp.* (Ce1-4) or *Lecane sp.* (Le1-4). Algae are in full dots and dashed curves – *M. minutum* and *C. vulgaris* in green, *C. reinhardtii* in light blue, *Cryptomonas sp.* in dark blue – and rotifers in empty dots and solid curves – *B. calyciflora* in dark red, *Cephalodella sp.* in red and *Lecane sp.* in orange.

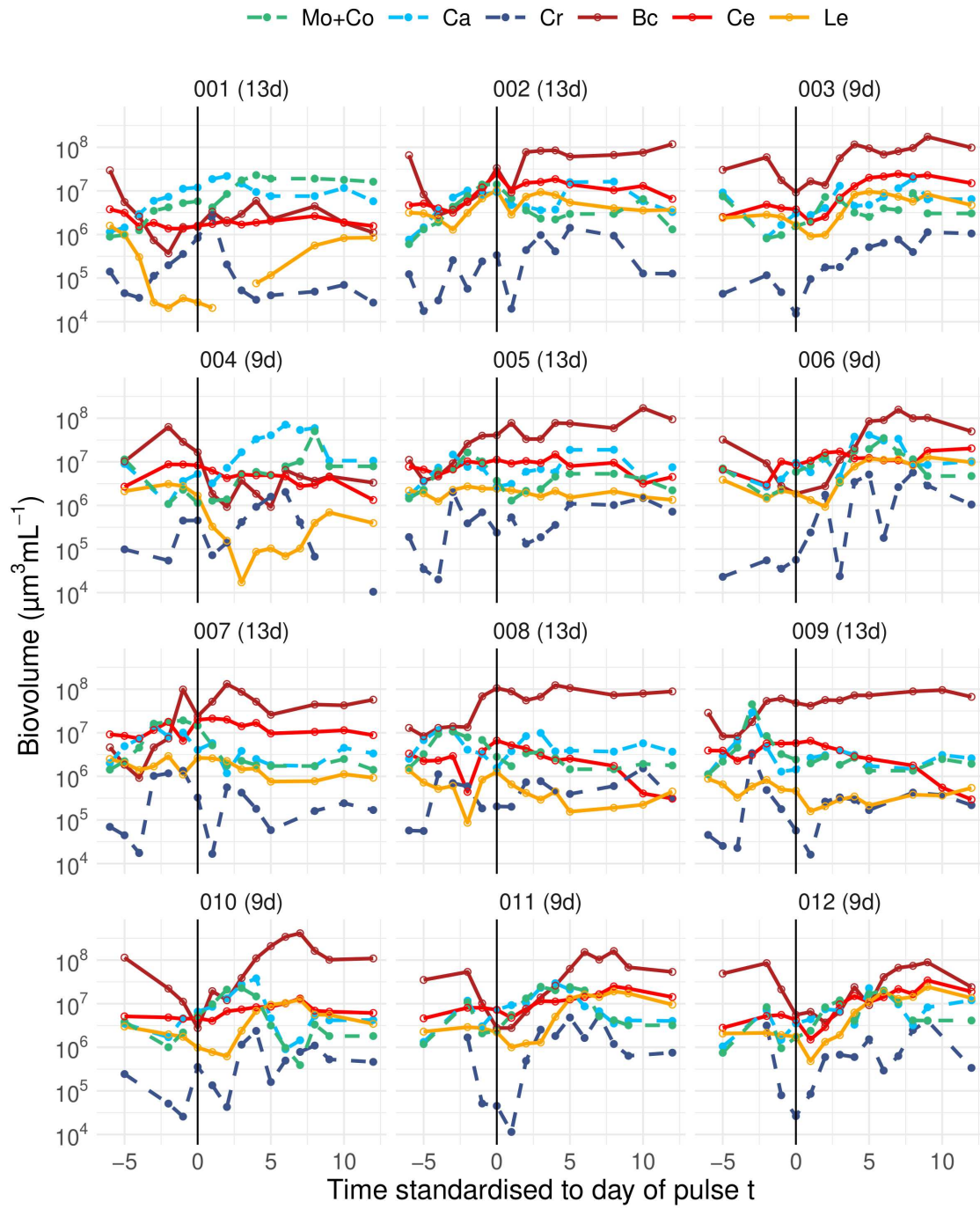


Figure S5.2: Biovolume timeseries of rotifer polycultures differing in the timing of the pulse (9d or 13d after inoculation) and thus TC values (cf. Table S5.1). Colours and markers are the same as in Figure S5.1.

Fitted biovolume timeseries

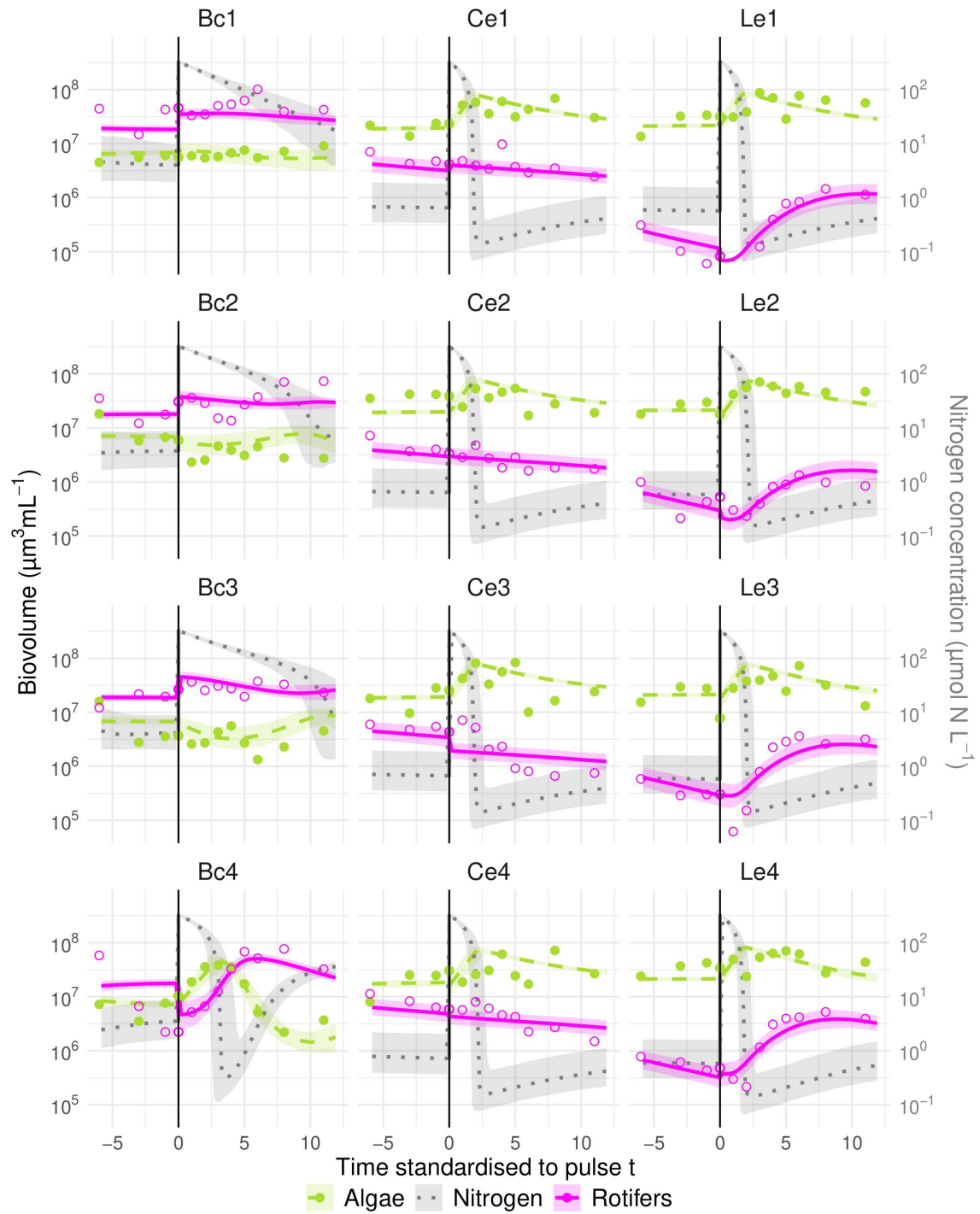


Figure S5.3: Biovolume timeseries of rotifer monocultures of *B. calyciflorus* (Bc1-4), *Cephalodella sp.* (Ce1-4) or *Lecane sp.* (Le1-4) fitted via Bayesian inference. Nitrogen concentration is in dotted grey curve, algal total biovolume is in dashed green curve and rotifer total biovolume is in solid magenta curve. The green empty circles and the magenta full circles are the experimental data of algae and rotifers, respectively.

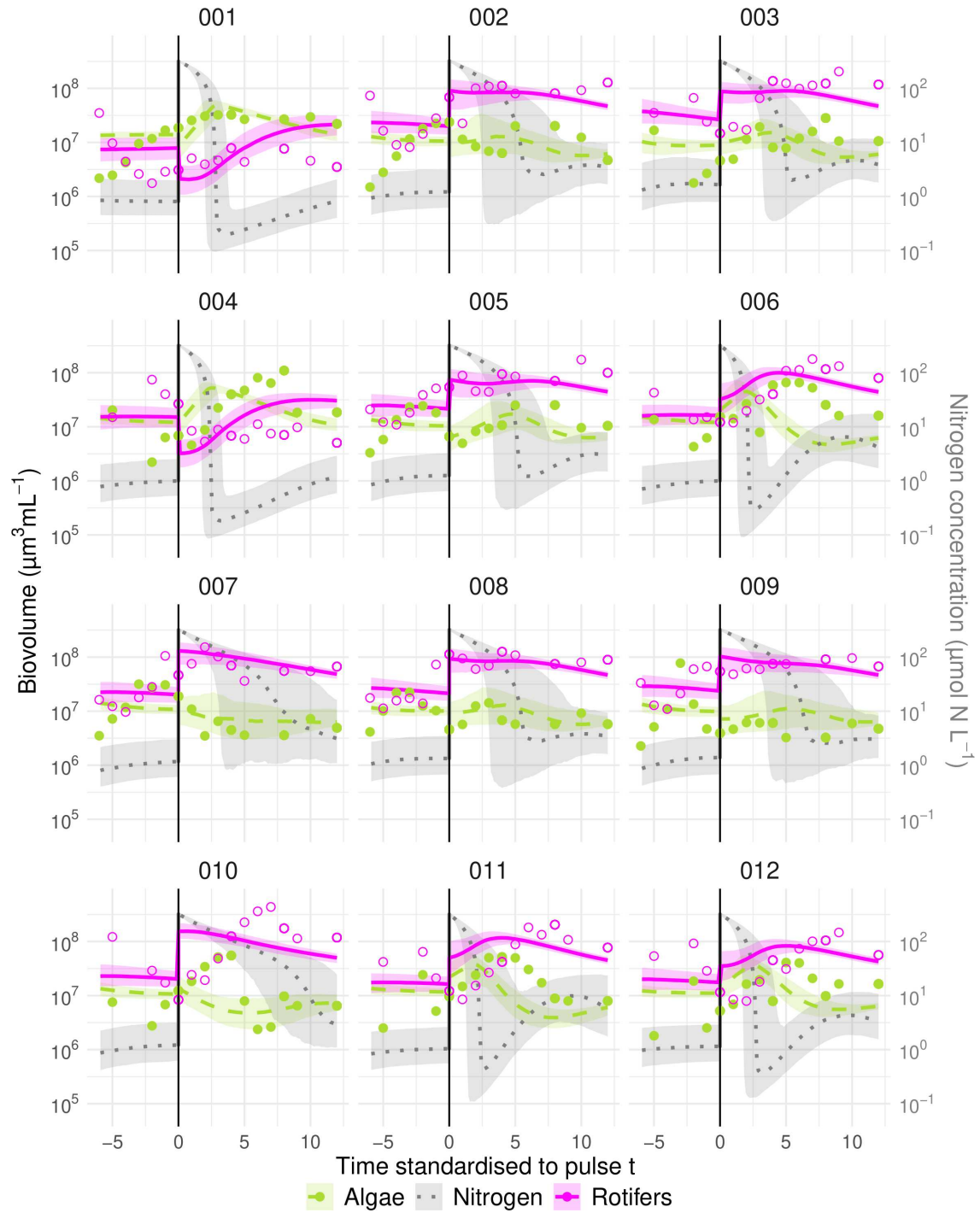


Figure S5.4: Biovolume timeseries of rotifer polycultures differing in the timing of the pulse (9d or 13d after inoculation) fitted via Bayesian inference. The figure layout is the same as in Figures 3 and S5.3.

Testing diversity effects on short-term responses to the pulse

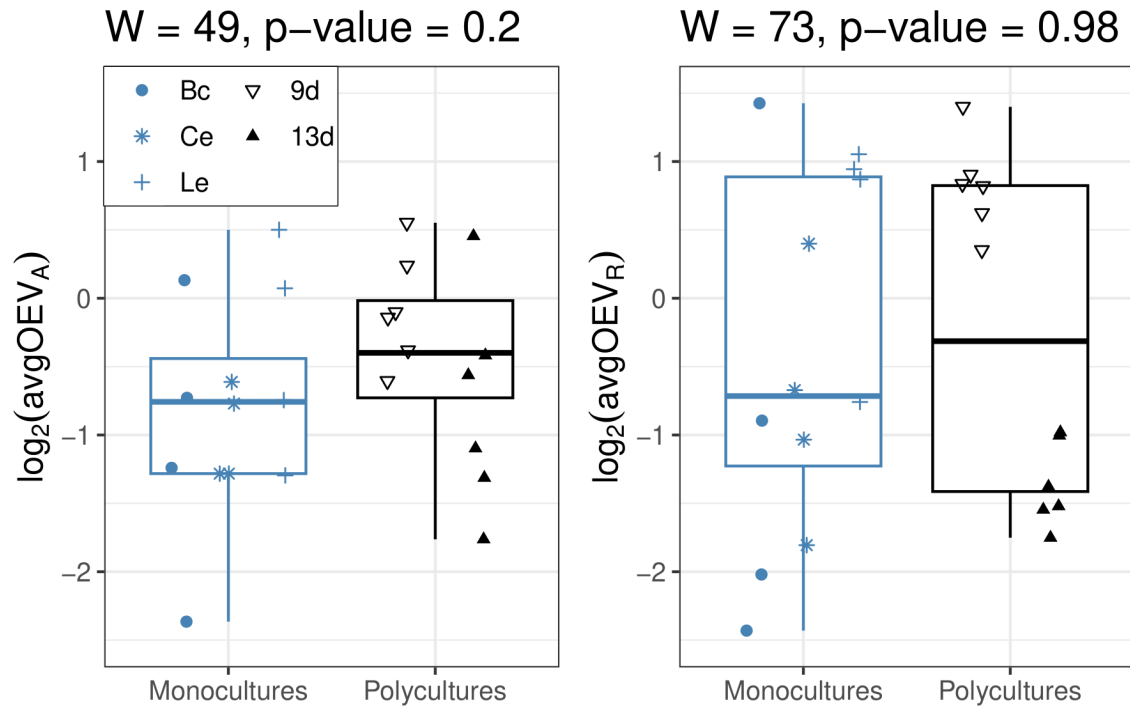


Figure S5.5: Differences in algal response (a) and rotifer response (b) between rotifer monocultures (low diversity, in blue) and rotifer polycultures (high diversity, in black). No significant differences were observed ($p\text{-value} > 0.05$ for all the Wilcoxon sum rank tests).

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